

ECN 594: Practice Midterm Exam

Instructions:

- You have **80 minutes**
- You may bring a calculator and notes on a two-sided cheat-sheet (letter-size paper)
- Please be neat. If your work is too messy it will not be graded.
- Be sure to show your working.
- This is a long exam, so there are lots of ways to get points. If you get stuck, move on!
- Good luck!

Name: _____

Question:	1	2	3	4	Total
Points:	30	30	20	20	100
Score:					

1. Short Answer Questions (30 points)

1. For each question, give a number, True/False/NEI, or a 1-2 sentence answer.
 - (a) (4 points) A monopolist faces demand with elasticity $\varepsilon = -4$ and has $MC = \$15$. What is the profit-maximizing price?
 - (b) (4 points) Consider the logit utility: $u_{ijt} = \beta \cdot \text{calories}_j + \alpha \cdot p_{jt} + \xi_{jt} + \varepsilon_{ijt}$. What is the economic interpretation of ξ_{jt} ?
 - (c) (4 points) True, False, or NEI: Adding demographic interactions to the logit model fully eliminates the IIA problem.
 - (d) (4 points) Why are cost shifters (e.g., input prices) valid instruments for price? Explain both relevance and exogeneity. (1-2 sentences)
 - (e) (4 points) True, False, or NEI: In the logit model, cross-price elasticity between products j and k depends on how similar the products are.
 - (f) (3 points) True, False, or NEI: Under marginal cost pricing ($P = MC$), a monopolist with fixed costs will earn zero profit.
 - (g) (4 points) A researcher claims that consumer surplus in logit is $CS = \delta_1 + \delta_2$. Consider a market with two goods and an outside option. Is this formula correct? Explain.

- (h) (3 points) True, False, or NEI: In a self-selection problem with two types, the high-type's incentive compatibility constraint always binds at the optimum.

2. Demand Estimation (30 points)

2. A researcher wants to estimate demand for smartphones using a logit model:

$$u_{ijt} = \beta \cdot \text{screen}_j + \alpha \cdot p_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

where screen_j is screen size (inches), p_{jt} is price, ξ_{jt} is unobserved product quality, and ε_{ijt} is i.i.d. Type 1 Extreme Value.

- (a) (6 points) Define the mean utility δ_{jt} . Using the Berry inversion, derive an expression for δ_{jt} in terms of observed market shares s_{jt} and the outside option share s_{0t} .

- (b) (6 points) Write down the linear regression equation the researcher would estimate to recover β and α . Clearly identify the dependent variable, regressors, and error term.

- (c) (6 points) Why does OLS produce biased estimates of α ? What is the likely direction of the bias? (2-3 sentences)

(d) (6 points) A researcher proposes using “cost shifters” (e.g., semiconductor chip prices) as instruments. Give one reason why these instruments might fail the exclusion restriction.

(e) (6 points) Compared to the basic logit, what is one advantage of adding demographic interactions (e.g., age $\times$ screen size) to the model?

3. Two-Part Tariffs (20 points)

3. A streaming service charges a monthly subscription fee F and a per-movie price p . Marginal cost per movie is $c = 1$.

There are 200 identical consumers, each with demand $q = 30 - 3p$ for movies per month.

- (a) (10 points) Write down the streaming service's profit as a function of F and p . (Hint: What is consumer surplus at price p ? What is the maximum F the service can charge?)

- (b) (10 points) Show that the profit-maximizing per-movie price is $p^* = c = 1$. (Hint: Take the derivative of profit with respect to p and set it equal to zero.)

4. Self-Selection (20 points)

4. An airline sells two ticket classes: Business (B) and Economy (E). There are two types of travelers: corporate travelers with high willingness to pay and leisure travelers with low willingness to pay.

Traveler Type	Willingness to Pay	
	Business Class	Economy Class
Corporate (50 travelers)	\$800	\$300
Leisure (100 travelers)	\$200	\$150

Marginal cost is \$50 for both classes.

- (a) (6 points) If the airline could perfectly identify traveler types, what prices would it charge for each class to each type? What is total profit?
- (b) (6 points) The airline cannot identify types but offers both classes. Write down the incentive compatibility (IC) constraint that ensures corporate travelers buy Business rather than Economy.

- (c) (8 points) Find the profit-maximizing prices for Business and Economy under self-selection. (Hint: Which IC constraint binds? Set it as an equality and solve.)